The internal energy of the Fermi gas at low temperatures is,

$$U = \sum_{\vec{k}} \frac{2\epsilon_{\vec{k}}}{e^{\beta(\epsilon_{\vec{k}}-\mu)}+1} = \frac{V}{(2\pi)^3}\int_0^\infty 4\pi k^2 dk \, \frac{2\hbar^2k^2/2m}{e^{\beta(\hbar^2k^2/2m-\mu)}+1}$$

$$= \frac{V\hbar^2}{2\pi^2 m}\int_0^\infty dk \left(\frac{1}{e^{\beta(\hbar^2k^2/2m-\mu)}+1}\right)k^4$$

$$= \frac{V\hbar^2}{2\pi^2 m}\left[\underbrace{\left(\frac{1}{e^{\beta(\hbar^2k^2/2m-\mu)}+1}\right)k^5/5\Big|_0^\infty}_{0} - \int_0^\infty dk\left(\frac{1}{e^{\beta(\hbar^2k^2/2m-\mu)}+1}\right)'\frac{k^5}{5}\right]$$

$$= -\frac{V\hbar^2}{2\pi^2 m}\int_0^\infty dk\left(\frac{1}{e^{\beta(\hbar^2k^2/2m-\mu)}+1}\right)'\frac{k^5}{5}$$

Again, since the derivative of the Fermi distribution function is sharply peaked near k_F.

$$-\left(\frac{1}{e^{\beta(\hbar^2k^2/2m-\mu)}+1}\right)' = \frac{e^{\beta(\hbar^2k^2/2m-\mu)}\,\beta\hbar^2k/m}{\left(e^{\beta(\hbar^2k^2/2m-\mu)}+1\right)^2}$$

$$\Rightarrow U \approx \frac{V\hbar^2}{10\pi^2 m}\frac{\beta\hbar^2}{m}\int_0^\infty \frac{e^{\beta(\hbar^2k^2/2m-\mu)}\,k^6}{\left(e^{\beta(\hbar^2k^2/2m-\mu)}+1\right)^2}dk$$

Let $\beta\left(\frac{\hbar^2k^2}{2m}-\mu\right) = x \Rightarrow \frac{\beta\hbar^2}{m}k\,dk = dx$

$k \to 0 \Rightarrow x \to -\beta\mu$

$k \to \infty \Rightarrow x \to \infty$

$k \approx k_F \Rightarrow |x| \ll 1$

$$k^2 = \frac{2m}{\beta\hbar^2}(x+\beta\mu)$$

$$\Rightarrow U \approx \frac{V\hbar^2}{10\pi^2 m}\int_{-\beta\mu}^\infty \frac{e^x}{(e^x+1)^2}\left(\frac{2m}{\beta\hbar^2}\right)^{5/2}(x+\beta\mu)^{5/2}dx$$

$$\approx \frac{V\hbar^2}{10\pi^2 m}\left(\frac{2m\mu}{\hbar^2}\right)^{5/2}\int_{-\infty}^\infty \frac{e^x}{(e^x+1)^2}\left(1+\frac{5}{2}\frac{x}{\beta\mu}+\frac{15}{8}\left(\frac{x}{\beta\mu}\right)^2+\cdots\right)dx$$

Since the integrand is sharply peaked near $x=0$, we have replaced the lower limit with $-\infty$.

$\dfrac{e^x}{(e^x+1)^2}$ is an even function of x

$$\Rightarrow U \approx \frac{V\hbar^2}{10\pi^2 m}\left(\frac{2m\mu}{\hbar^2}\right)^{5/2} 2\int_0^\infty \frac{e^x}{(e^x+1)^2}\left(1+\frac{15}{8}\left(\frac{x}{\beta\mu}\right)^2+\cdots\right)dx$$

Using $\displaystyle\int_0^\infty \frac{e^x}{(e^x+1)^2}dx = \frac{1}{2}$ and $\displaystyle\int_0^\infty \frac{e^x x^2}{(e^x+1)^2}dx = \frac{\pi^2}{6}$, we get,

$$U \approx \frac{V\hbar^2}{10\pi^2 m}\left(\frac{2m\mu}{\hbar^2}\right)^{5/2}\left[1+\frac{5\pi^2}{8}\left(\frac{T}{T_F}\right)^2\right]$$

$$\approx \frac{V\hbar^2}{10\pi^2 m}\left(\frac{2m}{\hbar^2}\right)^{5/2}\epsilon_F^{5/2}\left[1-\frac{\pi^2}{12}\left(\frac{T}{T_F}\right)^2\right]^{5/2}\left[1+\frac{5\pi^2}{8}\left(\frac{T}{T_F}\right)^2\right]$$

$$\approx \frac{V\hbar^2}{10\pi^2 m}\frac{2m}{\hbar^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\epsilon_F^{3/2}\,\epsilon_F\left[1+\left(-\frac{5}{24}+\frac{5}{8}\right)\pi^2\left(\frac{T}{T_F}\right)^2\right]$$

$$\approx \frac{V}{5\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} 3\pi^2 n\left(\frac{\hbar^2}{2m}\right)^{3/2}\epsilon_F\left[1+\frac{5\pi^2}{12}\left(\frac{T}{T_F}\right)^2\right]$$

$$= \frac{3}{5}N\epsilon_F\left[1+\frac{5\pi^2}{12}\left(\frac{T}{T_F}\right)^2\right]$$

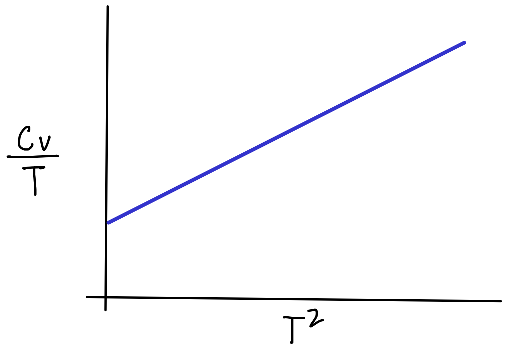
$$\Rightarrow C_V = \frac{\partial U}{\partial T} = \frac{3N\epsilon_F}{5}\times\frac{5\pi^2}{12}\frac{2T}{T_F^2} = \frac{\pi^2}{2}Nk_B\left(\frac{T}{T_F}\right)$$

The factor of T/T_F makes the electronic contribution to the specific heat of metals unimportant at room temperature.

The contribution to the specific heat from lattice vibrations is proportional to T^3.

Thus, $\quad C_V = \underbrace{AT}_{\text{electronic}} + \underbrace{BT^3}_{\text{lattice}}$

At extremely low temperatures, $AT \gg BT^3$ and the electronic contribution dominates. This happens below a temperature $\sqrt{A/B}$.

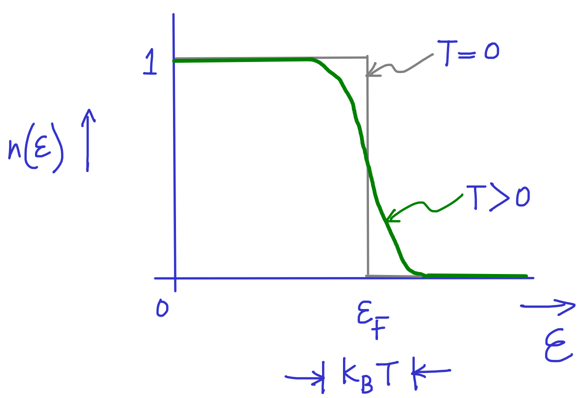


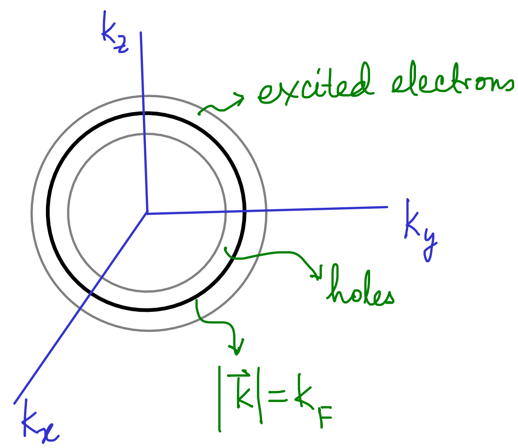
Why is the electronic contribution to the specific heat proportional to T?
At low temperatures, only a small fraction of electrons can be excited — they need to have an unoccupied site within an energy window comparable to k_BT. This is true for the electrons within an energy of about k_BT from the Fermi surface. The fraction of such electrons is about

$$\frac{k_BT}{\varepsilon_F} = \frac{T}{T_F}.$$

These electrons contribute k_BT towards the internal energy

$$\therefore U \sim k_BT \frac{NT}{T_F} \Rightarrow c_V \sim Nk_B\frac{T}{T_F}.$$





Fraction of excited electrons $\sim \dfrac{4\pi k_F^2\,\delta k}{\frac{4}{3}\pi k_F^3} \sim \dfrac{3\,\delta k}{k} \sim \dfrac{3}{2}\dfrac{\delta\varepsilon}{\varepsilon_F} \sim \dfrac{\delta\varepsilon}{\varepsilon_F} = \dfrac{T}{T_F}$ $\quad \because \varepsilon \propto k^2$

The actual numbers for an electron gas (discussed in the previous class)

$$n \approx 10^{22}/cm^3 = 10^{28}/m^3$$

$$\Rightarrow \varepsilon_F = \frac{\hbar^2}{2m}(3\pi^2 n)^{2/3}$$

$\hbar = 197\text{ MeV fm}/c$

$m = 0.51\text{ MeV}/c^2$

$$= \frac{(200\times10^6\times10^{-15})^2 \times (3\pi^2\times10^{28})^{2/3}}{2\times0.5\times10^6}\ eV$$

$$\Rightarrow \epsilon_F \approx \frac{4\times10^{-14}\times10\times5\times10^{18}}{10^6}\, eV \cdot = 2\, eV$$

$(3\pi^2)^{2/3} \approx 3^2 \approx 10$

$10^{2/3} = 100^{1/3} \approx 5$

$$T_F = \frac{\epsilon_F}{k_B} \approx \frac{2\times1.6\times10^{-19}}{1.38\times10^{-23}}\, K \approx 2\times10^4\, K.$$

At low but non-zero temperatures ($T \ll T_F$) some electrons below the Fermi level get excited and acquire energies that are higher than ϵ_F. This creates "holes" below the Fermi level.

At room temperature, fraction of excited electrons $\sim \frac{300}{20000} = 1.5\%$

Transport in Metals (Electrical and Thermal conduction)

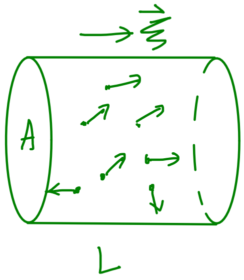
Results from classical arguments (Drude theory):

In Drude theory electrons in a metal are treated as a classical ideal gas. In the absence of an external electric field, electrons move randomly in all possible directions and have zero average velocity. They collide with each other and after each collision acquire a random velocity that is consistent with the Maxwell-Boltzmann distribution. When an electric field is applied, the electrons are accelerated between collisions in a direction parallel to the electric field. The average velocity, therefore becomes non-zero. When the electric field is not too large, this average velocity (known as the **drift velocity**) is proportional to the applied electric field.

$v = \mu \xi$, where ξ is the electric field (E is not used since we will use it for the energy).

The constant of proportionality is known as the **mobility** of the electrons. The resulting current density, J, is also proportional to the electric field, the constant of proportionality being the electrical conductivity, σ.

$$J = \sigma \xi$$



Consider a cylinder with cross section perpendicular to the electric field, $\vec{\xi}$.

If electrons, on average cover the length of the cylinder in time t, then, the current flowing across the cross section of the cylinder per unit time is,

$$I = \frac{(nAL)e}{t} = evAn,$$

(Here, we are only writing down the magnitude of I)

where n = eletron number density (number of electrons per unit volume)

v = average velocity.

$$\Rightarrow J = \frac{I}{A} = nev$$

$$\Rightarrow \sigma = ne\mu$$

Mean free time, τ = average time between collisions.

During this time, The electron acquires a momentum,

$$mv_d = e\xi\tau$$

$\Rightarrow$ The mobility, $\mu = \frac{v_d}{\xi} = \frac{e\tau}{m}$

$$\Rightarrow \boxed{\sigma = \frac{ne^2\tau}{m}}$$

Thermal Conduction

From Fourier's law, the heat current, $\vec{J}^q$ is proportional to the negative of the temperature gradient.

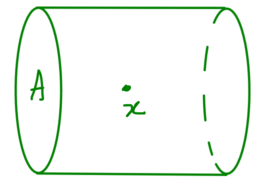
$\vec{J}_q = -K\nabla T$, where K is the thermal conductivity.

J_q is the thermal energy flowing across unit area per unit time.

If the temperature gradient is only in the x-direction, $J_q = -K\frac{dT}{dx}$

The electrons reaching x may have come from the low or high temperature side, in either case, their speeds/energies are randomly chosen at the time of their last collision, which must have occured, on average, at a time τ earlier. Thus, the electrons, on average, have energies, $E(x \pm v_x\tau)$. Of the electrons that come from $x + v_x\tau$, only half move towards x (with speed v in the $-x$ direction). The same is true for the electrons that come from $x - v_x t$.

The number of electrons that cross unit area around x per unit time is $\frac{nAv_x t}{At} = nv_x$



∴ The thermal current at x is

since these electrons move in the opposite direction

$$J_q = \frac{1}{2} n v_x E(x - v_x\tau) - \frac{1}{2} n v_x E(x + v_x\tau)$$

$$= \frac{1}{2} n v_x \left[E(x) - v_x\tau \frac{dE}{dx} - E(x) - v_x\tau \frac{dE}{dx} \right]$$

$$= -n v_x^2 \tau \frac{dE}{dx}$$

$$= -n v_x^2 \tau \frac{dE}{dT}\frac{dT}{dx}$$

$$= -\frac{1}{3} c_v v^2 \tau \frac{dT}{dx}$$

$v_x^2 = \frac{1}{3} v^2$,

$n\frac{dE}{dT} = c_v$ = heat capacity per unit volume

$$\Rightarrow \quad K = \frac{1}{3} c_v v^2 \tau$$

$$\therefore \quad \frac{K}{\sigma T} = \frac{c_v m v^2}{3ne^2 T} \equiv \text{Lorenz number}$$

For a classical ideal gas (Drude theory of metals),

$$\frac{1}{2}mv^2 = \frac{3}{2}k_B T \text{ and } c_v = \frac{3}{2} n k_B \Rightarrow \frac{K}{\sigma T} = \frac{3}{2}\left(\frac{k_B}{e}\right)^2 = 1.11 \times 10^{-8} \, (V/K)^2$$

For a Fermi gas (Sommerfeld theory of metals), at low temperature, the electrons which participate in thermal and electrical conduction lie near the Fermi surface.

$$\therefore \quad v^2 \approx v_F^2 \Rightarrow \frac{1}{2}mv^2 \approx \varepsilon_F$$

$$c_v = \frac{C_v}{V} \approx \frac{\pi^2}{2}\frac{nk_B T}{T_F} = \frac{\pi^2}{2}\frac{nk_B^2 T}{\varepsilon_F}$$

$$\Rightarrow \frac{K}{\sigma T} \approx \frac{C_v m v^2}{3ne^2 T} \approx \frac{\pi^2}{2}\frac{n k_B^2 T}{\varepsilon_F} 2\varepsilon_F \frac{1}{3ne^2 T} = \frac{\pi^2}{3}\left(\frac{k_B}{e}\right)^2 = 2.44\times 10^{-8} \left(V/K\right)^2$$

The above relation is known as the **Wiedemann-Franz law**,

$$\boxed{\frac{K}{\sigma} = \frac{\pi^2}{3}\left(\frac{k_B}{e}\right)^2 T}$$

This approximately holds true for many metals at room temperature.